

## **Drilling & Blasting Interview Guidelines**

### **Drill Machineries Specification:**

#### **F-9 =>**

- Type of Engine : C-9 DI
- Output Power : 224 Kw
- Track Speed : 3.6 Km/H
- Ground Clearance : 405mm
- Drill Bit Diameter : 115mm
- Noise Level (Inside) : 82.5 dBA
- Idle Speed : 109.1 dBA
- Maximum Speed : 117 dBA
- Drill Speed : 122.9 dBA
- Hydraulic Oil (SS46) : 300 Ltr
- Compressor Oil (HL68) : 38 Ltr
- Track Gear Oil (HP30) : 3 x 2.6 Ltr
- Engine Oil (Cat D) : 35 Ltr
- Diesel Capacity : 400 Ltr

#### **L-8 =>**

- Type of Engine : CAT C-13
- Output Power : 402 Kw
- Track Speed Maximum : 3.2 Km/H
- Ground Clearance : 405mm
- Drill Bit Diameter : 165mm
- Noise Level (Inside) : 85 dBA
- Idle Speed : 115.5 dBA
- Drill Speed : 125 dBA
- Hydraulic Oil (SS46) : 380 Ltr
- Compressor Oil (HL68) : 63 Ltr
- Track Gear Oil (HP30) : 3 x 2.6 Ltr
- Diesel Capacity : 760 Ltr

- Air Oil : 20 Ltr
- Maximum Air Pressure : 25 BAR
- Type of Compressor : XRX-10
- Track Oscillation (F/R) :  $10^0/10^0$
- Weight : 22600Kg

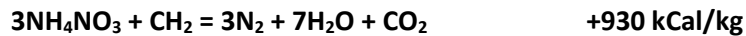
## **Blasting Practice**

**Ammonium Nitrate(AN):** 160mm Dia  
1m = 17-18Kg require in L-8 hole.

**ANFO :** Prilled ammonium nitrate (AN) and Fuel Oil (FO) known as ANFO.

- **USED:** ANFO mixtures now account about 70% of the total explosives used in Iron & Lime Stone Mines.

- When 94% (AN) & 6% (FO) are mixed the following reaction taking:-



**Bulk ANFO:** It is suitable for wet holes. The main reason being Ammonium Nitrate will observe water.

**Density of Bulk Explosive:**

- ⇒ Watery Hole – 1.1gm/cc
- ⇒ Dry Hole – 0.8-0.9gm/cc

**Classifications of Explosive & Accessories: -**

**CLASS- 1: - Gun Power**

**CLASS- 2: - Nitrate Mixture (ANFO)**

**CLASS- 3: - Nitro Compound (Cast Booster)**

**CLASS- 4: - Chlorate Mixture**

**CLASS- 5: - Fulminate**

**CLASS- 6: - Ammunition Safety Fuse/ Detonating Fuse**

**CLASS- 7: - Fireworks**

**CLASS- 8: - LOX (Liquid Oxygen Explosive)**

**Stemming Require:** 50-60% depth of hole.

**Safety Fuse:** Class – 6 & Division – 1.

**Detonating Fuse:** Class – 6 & Division – 2.

**Detonators (Nonel):** Class – 6 & Division - 3

- **TLD (Trunk Line Delay Detonator):** 17ms, 25ms, 42ms, 65ms.
- **Long Delay** –  $\frac{1}{2}$  Second delay= 300ms
- **Short Delay** – (0 to 6) 25ms, 7 & 8 50ms, 9 & 10 75ms

- DTH – 200ms & 250ms

**Explosives: Nitrate Mixture (Emuldyn-IDL & Powergel – Orica): Class -2 & Division- 0.**

**Type of Explosive - Slurry Explosive (Powergel Nitrate Mixture)**

- 25mm (1 Cartridge) = 125gm
- 40mm (1 Cartridge) = 390gm
- 83mm (1Cartridge) = 2.78kg
- 125mm (1 Cartridge) = 6.25kg
- Boost = 250gm

**In L-8 (165mm Dia) Holes:**

- In 4m Depth
  - ⇒ AN – 12-15kg.
  - ⇒ Exp. – 1 Cartridge of 83mm (2.78kg)
- In 6m Depth:
  - ⇒ AN = 22-25kg
  - ⇒ Exp. – 1 Cartridge of 83mm (2.78kg)
- In 8m Depth:
  - ⇒ AN = 60-70kg
  - ⇒ Exp. – 1 Cartridge of 125mm(6.25kg) **or** 1 Booster (250gm)
- In 10m Depth:
  - ⇒ AN = 80kg
  - ⇒ Exp. – 1 Cartridge of 125mm(6.25kg) **or** 1 Booster (250gm)
- In 12m Depth:
  - ⇒ AN = 90-120kg
  - ⇒ Exp. – 2 Cartridge of 125mm(12.50kg) **or** 1 Booster (250gm)

**In F-9 & Sandvik Holes:**

- Horizontal Holes:
  - ⇒ AN = 2 Cartridge of 90mm (2 x 2.5 kg = 5kg).
  - ⇒ Exp. – 1 Cartridge of 40 mm (400gm).
- Vertical Holes:
  - ⇒ AN = 3 Cartridge of 90mm (3 x 2.5 kg = 7.5kg).
  - ⇒ Exp. – 1 Cartridge of 40 mm (400gm).

**Charge Factor maintained as per CMFRI = 0.24 to 0.42 kg/m<sup>3</sup>**

$$\text{Charge Factor} = \frac{\text{Total Explosive}}{S \times B \times H \times \text{No. of holes}} = \frac{\text{Total Explosive}}{\text{m}^3} \text{ (Kg/m}^3\text{)}$$

$$\text{Powder Factor} = \frac{S \times B \times H \times \text{No. of holes} \times \text{Sp. gr.}}{\text{Total Explosive}} \quad (\text{PF} = 7-10 \text{ Te/kg})$$

**Blasting Tools:**

- ⇒ Series connection ⇒ Dynamo Exploder
- ⇒ Seismograph is used to take Vibration.
- ⇒ PPV : Past Particle Velocity.

**Safety Device of PC's:**

- ⇒ Swing lock,
- ⇒ Fire resistance type
- ⇒ Canopy
- ⇒ Audio-visual Alarm

**Bucket Capacity:** 3.5m<sup>3</sup> and 6 Cylinder

**Production in a Day:**

**At Jaypee Cement : Baga (H.P.)**

Crusher = 900 TPH  
Row Material = 15000 TPD  
TPD Clinker = 10,000 TPD  
6500 TPD

**At Ambuja Cement: Darlaghat (H.P.)**

- Crusher = 800 TPH  
- ROW Material = 24000  
- Clinker (Suli) =

(Rauri) = 7500 TD

**Capacity of Plant: (Total 4.0 MT)**

SULI PLANT : 2.2 MT  
RAURI PLANT: 1.8 MT  
ROW material use for per ton of Clinker produce = 1.64 te.

**Drilling Rate:**

○ By L-8:

- ⇒ 40-50mtrs/hr ( In Hard Rock) ⇒ 30-35mtrs/hr (In Soft Rock).
- ⇒ Drilling Cast: Rs. 3000/- per hrs of 30mtr drilling.

○ F-9 & Sandvik:

- ⇒ 35-40mtrs/hr

**Cost of Machineries**

**In per ton of Lime stone production: - (Rs. 105-110)**

|                |   |             |
|----------------|---|-------------|
| Salary & wages | – | Rs. 5/te    |
| Volvo          | – | Rs. 12/te   |
| Drilling       | – | Rs. 4/te    |
| Excavator      | – | Rs. 11/te   |
| Royalty        | – | Rs. 63/te   |
| Blasting       | – | Rs. 3-4/te  |
| Miscellaneous  | – | Rs. 7-11/te |

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Total = Rs.105-110/te

**Formula of lime stone quality maintained: - (QL)**

$$QL = \frac{Q_1L_1 + Q_2L_2 + Q_3L_3}{L_1 + L_2 + L_3} \quad \begin{array}{l} Q - \text{Quantity loaded by particular excavators} \\ L - \text{LSF of loaded material by particular excavators} \end{array}$$

**Lime saturation factor (LSF): -**

$$LSF = \frac{\text{CaO}}{2.8 \text{ SiO}_2 + 1.18 \text{ Al}_2\text{O}_3 + 0.65 \text{ Fe}_2\text{O}_3} \times 100\%$$

- ✦ Free lime should not be more than 1.5 %.
- ✦ After burning in kiln lime percentage increasing around 1.5% (e.g. 42% to 63%).
- ✦ Magnesia not more than 3%.

**Excavator Specification: -**

- Bucket capacity : 3.5m<sup>3</sup>
- Cycle time of excavator : 45 sec.
- No. of passes : 5
- Bucket fill factor : 0.8%
- Specific gravity of lime stone (Before blasting or in-situ) : 2.5
- Specific gravity of lime stone (After blasting) : 1.5
- Soil factor (Before / After blasting) : 2.5/1.5 = 1.7

## **Returns Maintained By Office**

### **IBM =>**

✚ Monthly Returns => a) Form F-1  
b) Form – N

✚ Annual Returns => Form H-1

### **Police Station (To SHO of Near Police Station) =>**

15<sup>th</sup> day Returns (Forth Nightly Report) => Explosives, Detonator's.

### **Deputy Chief Controller of Explosive=>**

Monthly & Quarterly Returns =>

- a) R-9 => Ammonium Nitrate (AN) Stock: In & Out (Used).
- b) RE-7 => Explosives Stock: In & Out (Used).

Quarterly Returns=> Used

### **DGMS**

- ✚ Annual Returns: First Schedule Form –III (Regulation-5)
- ✚ Quarterly Returns: First Schedule Form – II (Regulation-4)